Day 8 – Access Point, Repeater and Bridge

Aim

To understand the working of Access Point, Repeater, and Bridge, and to simulate a wireless network using Cisco Packet Tracer.

Objective

- To understand the purpose of an Access Point.
- To understand how a Repeater extends wireless coverage.
- To understand how a Bridge connects two LAN segments.
- To build and configure a simple wireless network in Cisco Packet Tracer.
- To verify communication between a wired PC and a wireless laptop.

Devices Used

- 1 PC
- 1 Laptop
- 1 Access Point
- 1 Switch
- Copper Straight-Through Cable
- Cisco Packet Tracer

Network Topology

Laptop

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Access Point

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Switch

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PC0

Procedure

Step 1

Open Cisco Packet Tracer.

Step 2

Add the following devices:

- 1 PC
- 1 Laptop
- 1 Switch
- 1 Access Point

Step 3

Connect the devices.

PC0 → Switch (Copper Straight-Through)

Access Point → Switch (Copper Straight-Through)

Laptop → Wireless Connection

Step 4

Configure the Access Point.

SSID:

College_WiFi

Step 5

Install the wireless module (WPC300N) in the laptop.

- Turn OFF the laptop.

- Remove the Ethernet module (if present).
- Install the WPC300N wireless module.
- Turn ON the laptop.

Step 6

Connect the laptop to the wireless network.

Desktop → PC Wireless

Select:

College_WiFi

Click **Connect**.

Step 7

Assign IP addresses.

PC0

IP Address : 192.168.1.2

Subnet Mask: 255.255.255.0

Laptop

IP Address : 192.168.1.3

Subnet Mask: 255.255.255.0

Step 8

Test connectivity.

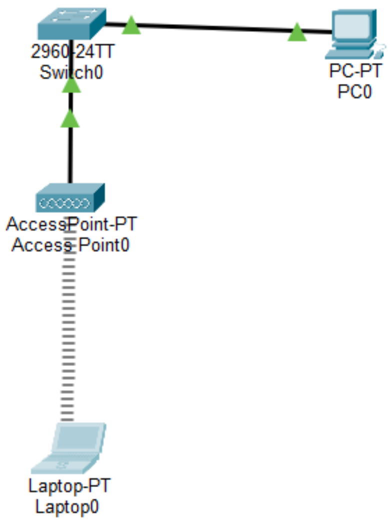
Open Command Prompt on the Laptop.

Execute:

ping 192.168.1.2

Successful replies indicate proper communication between the wireless laptop and the wired PC.

Screenshot



Observation

- The Access Point successfully provided wireless connectivity to the laptop.
- The laptop connected to the configured SSID (College_WiFi).
- Communication between the wired PC and wireless laptop was successful.
- The Access Point acted as a bridge between the wireless and wired network.

Result

The wireless network was successfully created using Cisco Packet Tracer. The laptop connected to the Access Point wirelessly and successfully communicated with the wired PC through the switch.

Conclusion

This experiment demonstrated the role of an Access Point in connecting wireless devices to a wired LAN. It also reinforced the concepts of wireless communication, IP addressing, and connectivity testing using the ping command. The experiment provided a practical understanding of how Wi-Fi networks operate in homes, colleges, and organizations.